



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AC BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

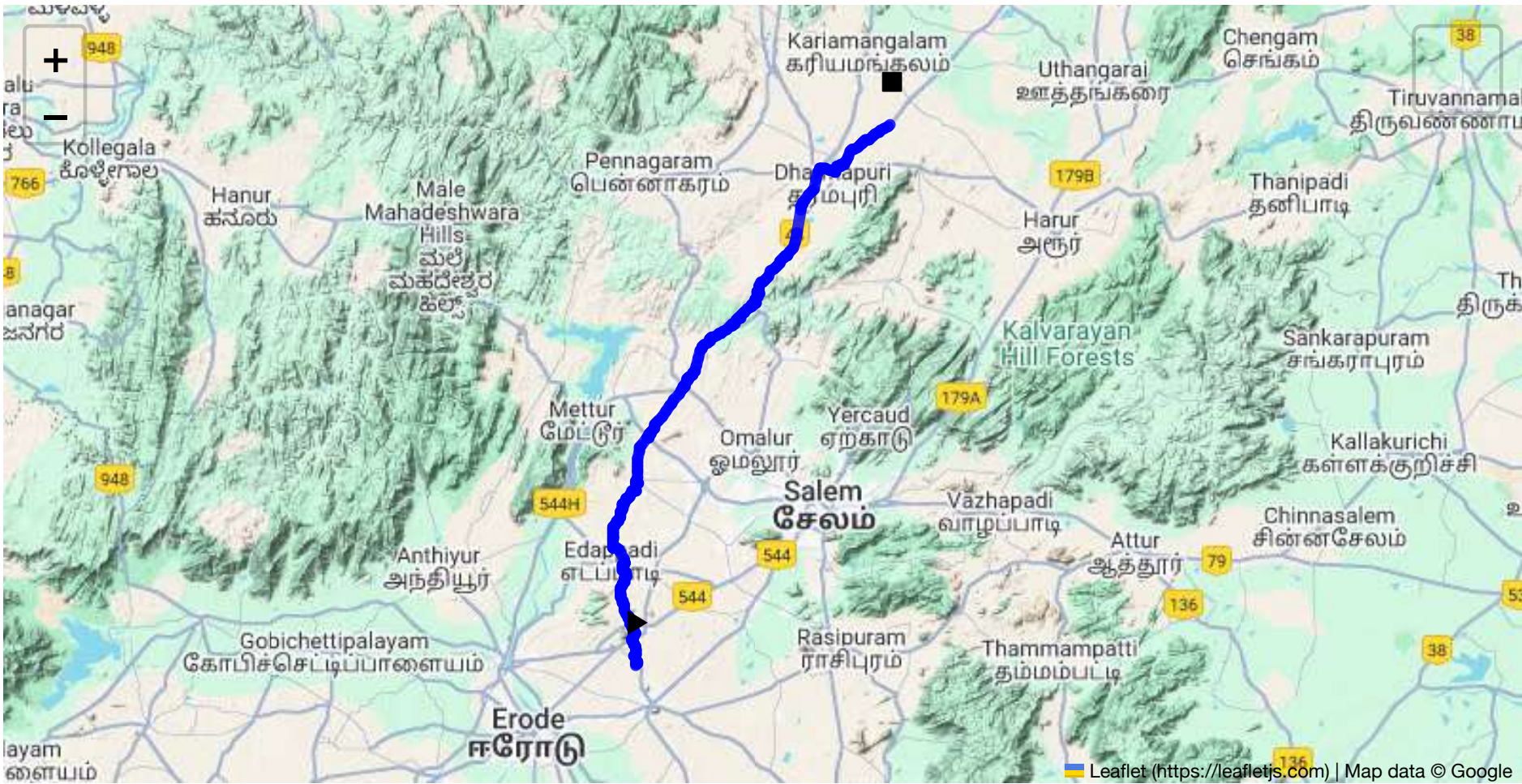
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 108.89 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours

Start: (11.4381, 77.8734)
End: (12.203365, 78.2425)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route from Sangagiri to Andihalli via Kottai and Dharmapuri spans approximately 108.89 kilometers, primarily along state highways and national highways. It serves as an important connector across rural and semi-urban areas of Tamil Nadu, featuring a mix of straight highways and winding rural roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- Weather Conditions:** The region typically experiences a tropical climate, with high temperatures in the summer and moderate rainfall during the monsoon season (June to September).
- Hazards:** Monsoon rains can lead to water-logging and reduce road traction, increasing the risk of skidding, particularly in rural segments. Visibility can also be affected during heavy rains.

3. Traffic Patterns

- Peak Hours:** Traffic density generally rises in the morning (7:30 AM - 9:30 AM) and evening (6:00 PM - 8:00 PM) as people travel to and from work.
- Congestion-Prone Areas:** Key congestion points are around major towns like Edappadi and Dharmapuri, where local traffic merges with highway traffic.

4. Assessment of Road Quality and Infrastructure

- Road Quality:** The condition is variable; highways are generally well-maintained but rural roads may have potholes and uneven surfaces.

- **Infrastructure:** Limited roadside amenities in rural stretches. Petrol stations and emergency services are more frequent near urban areas.

5. Alternative Routes for Emergencies

- In case of road blockages, a potential detour could be via the Salem-Orikkai route, though this adds extra distance and is only advisable in significant emergencies.

6. Local Regulations for Hazardous Material Transport

- Transporting hazardous materials requires specific permits. Vehicles must be properly labeled, and drivers should possess training certificates on handling dangerous goods.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- There have been sporadic reports of trucks overturning due to sharp curves in rural segments, mainly attributed to speeding or adverse weather conditions.
- No major incidents involving hazardous materials have been recorded in recent years, but caution is always advised.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Watch for wildlife crossings in rural stretches, particularly at night.
- **Environmental Impact:** Ensure minimal disturbance to local flora and fauna by adhering to designated routes and speed limits.

9. Communication Coverage

- There are potential communication dead zones, especially in rural and hilly areas between Edappadi and Dharmapuri.
- Ensure alternative communication means, like satellite phones, are available for emergency uses.

10. Estimated Emergency Response Times

- **Urban Segments:** Estimated response within 30-45 minutes.
- **Rural Segments:** Could be longer, approximately 60-90 minutes, due to diverse terrain and lower presence of emergency services.

11. Overall Summary of Risk Assessment

- **High Risks:** Weather-related road conditions during the monsoon, potential communication challenges, and road quality in rural areas.
- **Medium Risks:** Traffic congestion during peak hours, and limited infrastructure in certain stretches.
- **Low Risks:** Regulatory compliance if all requirements are adhered to reliably.

In conclusion, while the route presents some challenges due to variable road conditions and potential weather disruptions, careful planning and adherence to safety regulations can mitigate most significant risks. Preparing adequately for communication and emergency scenarios will ensure a safer journey for vehicles transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Roundabout	High	12.13658, 78.16171	15 KM/Hr
3	Roundabout	High	12.14589, 78.17156	15 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	High	11.44900, 77.87406	15 KM/Hr
7	Turn	High	11.45028, 77.87202	15 KM/Hr
8	Turn	Medium	11.45309, 77.87325	30 KM/Hr
9	Turn	Medium	11.46400, 77.87323	30 KM/Hr
10	Turn	Medium	11.46765, 77.87037	30 KM/Hr
11	Turn	High	11.47403, 77.86841	15 KM/Hr
12	Turn	Medium	11.47468, 77.86913	30 KM/Hr
13	Turn	Medium	11.47934, 77.87008	30 KM/Hr
14	Turn	Medium	11.48045, 77.87147	30 KM/Hr
15	Turn	High	11.48277, 77.87250	15 KM/Hr
16	Turn	Medium	11.49496, 77.86660	30 KM/Hr
17	Turn	Medium	11.49835, 77.86590	30 KM/Hr
18	Turn	Medium	11.50914, 77.86023	30 KM/Hr
19	Turn	Medium	11.52571, 77.85522	30 KM/Hr
20	Turn	Medium	11.52772, 77.85585	30 KM/Hr
21	Turn	Medium	11.54729, 77.85264	30 KM/Hr
22	Turn	Medium	11.55650, 77.85798	30 KM/Hr
23	Turn	Medium	11.56401, 77.85969	30 KM/Hr
24	Turn	High	11.57893, 77.85430	15 KM/Hr
25	Turn	Medium	11.58000, 77.85722	30 KM/Hr
26	Turn	Medium	11.58700, 77.85827	30 KM/Hr
27	Turn	Medium	11.60952, 77.84153	30 KM/Hr
28	Turn	Medium	11.62044, 77.84101	30 KM/Hr
29	Turn	Medium	11.64049, 77.84596	30 KM/Hr
30	Turn	Medium	11.64166, 77.84871	30 KM/Hr
31	Turn	Medium	11.64578, 77.85139	30 KM/Hr
32	Turn	Medium	11.66440, 77.85591	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
33	Turn	Medium	11.67281, 77.86013	30 KM/Hr
34	Turn	Medium	11.67398, 77.86254	30 KM/Hr
35	Turn	High	11.68487, 77.86888	15 KM/Hr
36	Turn	High	11.68478, 77.87285	15 KM/Hr
37	Blind Spot	Blind Spot	11.69463, 77.87589	10 KM/Hr
38	Turn	Medium	11.69531, 77.87507	30 KM/Hr
39	Turn	High	11.69549, 77.87444	15 KM/Hr
40	Turn	High	11.76020, 77.89071	15 KM/Hr
41	Blind Spot	Blind Spot	11.75960, 77.89221	10 KM/Hr
42	Turn	Medium	11.83107, 77.94362	30 KM/Hr
43	Turn	Medium	11.83228, 77.94376	30 KM/Hr
44	Turn	Medium	11.92962, 78.02347	30 KM/Hr
45	Turn	High	11.92961, 78.02392	15 KM/Hr
46	Turn	High	11.95066, 78.04616	15 KM/Hr
47	Turn	High	11.95048, 78.04668	15 KM/Hr
48	Turn	High	11.95084, 78.04711	15 KM/Hr
49	Turn	High	11.95818, 78.04978	15 KM/Hr
50	Turn	Medium	11.95988, 78.04948	30 KM/Hr
51	Turn	Medium	11.96304, 78.05136	30 KM/Hr
52	Turn	High	11.96399, 78.05135	15 KM/Hr
53	Turn	High	11.96580, 78.04923	15 KM/Hr
54	Turn	Medium	11.96675, 78.04939	30 KM/Hr
55	Turn	Medium	11.96791, 78.05070	30 KM/Hr
56	Turn	Medium	12.14026, 78.13955	30 KM/Hr
57	Turn	Medium	12.14048, 78.13998	30 KM/Hr
58	Turn	High	12.14499, 78.16500	15 KM/Hr
59	Turn	Medium	12.14701, 78.17527	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
9	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
10	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
11	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
12	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
13	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
15	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
16	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
17	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
18	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
19	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
20	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium
21	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
22	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
23	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
24	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
25	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
26	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
28	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
29	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
30	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
31	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
32	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
33	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
34	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
35	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
36	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13658, 78.16171



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14589, 78.17156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



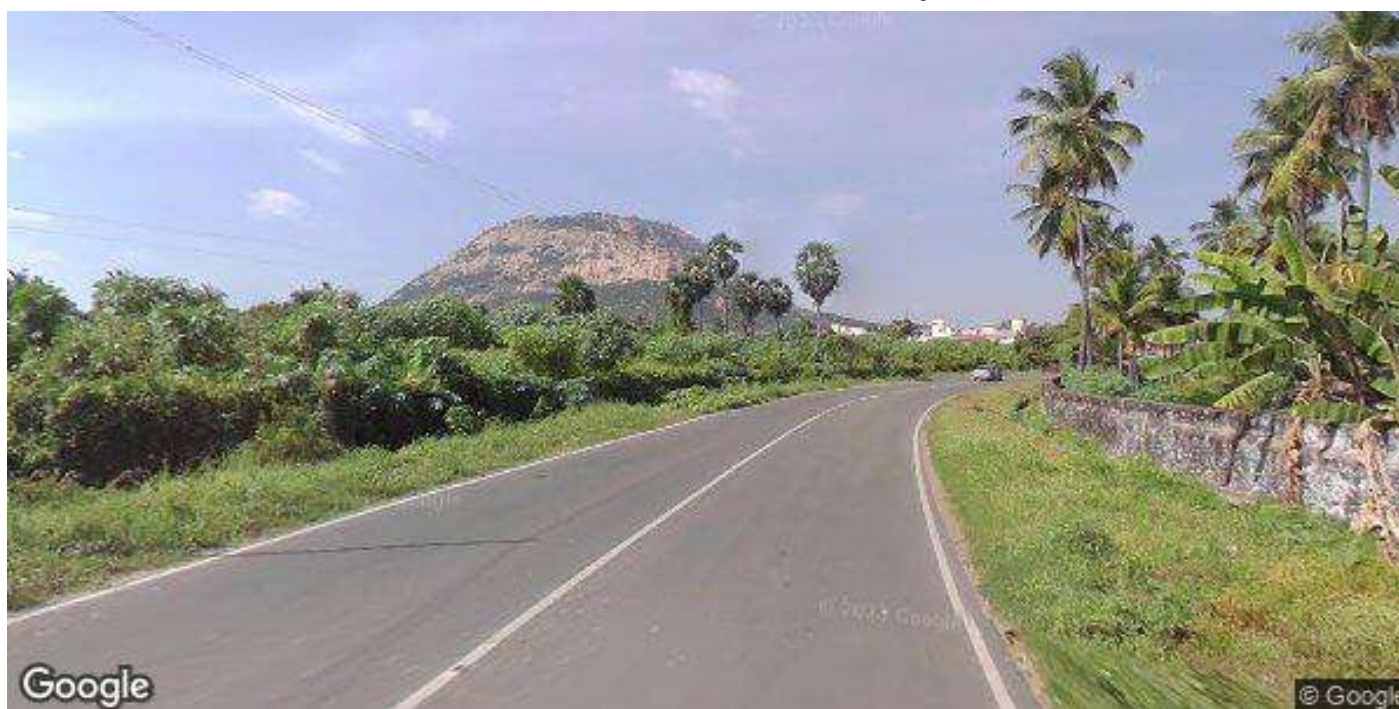
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47468, 77.86913



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47934, 77.87008



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48045, 77.87147



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48277, 77.87250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49496, 77.86660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49835, 77.86590



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50914, 77.86023



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52772, 77.85585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54729, 77.85264



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55650, 77.85798



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56401, 77.85969



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58000, 77.85722



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58700, 77.85827



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60952, 77.84153



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62044, 77.84101



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64049, 77.84596



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64166, 77.84871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64578, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66440, 77.85591



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67281, 77.86013



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67398, 77.86254



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68487, 77.86888



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68478, 77.87285



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69463, 77.87589



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69531, 77.87507



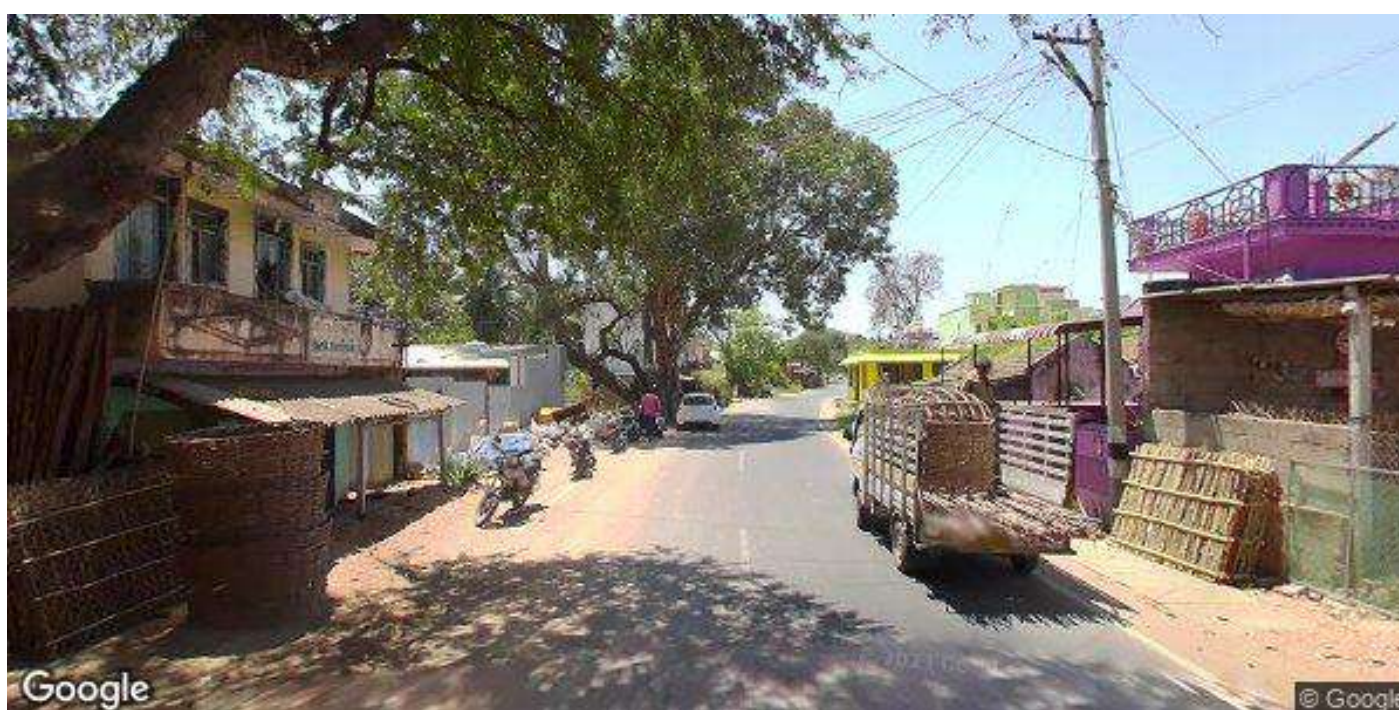
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69549, 77.87444



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76020, 77.89071



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.75960, 77.89221



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83107, 77.94362



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83228, 77.94376



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95066, 78.04616



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95048, 78.04668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95084, 78.04711



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14026, 78.13955



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14048, 78.13998



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14499, 78.16500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14701, 78.17527

